

THE NONLINEAR DYNAMICS OF INTERNET MEMES

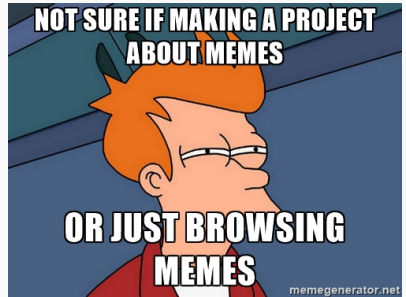
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INTRODUCTION

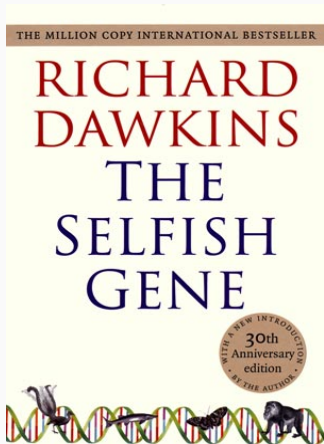
A meme is "an idea, behavior, or style that spreads from person to person within a culture"

An Internet meme (a subset of the general concept) is an activity, concept, catchphrase or piece of media which spreads, often as mimicry, from person to person via the Internet

The concept of a meme was defined as an analogue to the *gene*, where it could undergo phenomena such as variation, mutation, competition, and inheritance



Websites like memegenerator.net make it really easy to create and post memes



"We need a name for the new replicator, a noun that conveys the idea of a unit of cultural transmission, or a unit of imitation. 'Mimeme' comes from a suitable Greek root, but I want a monosyllable that sounds a bit like 'gene'. I hope my classicist friends will forgive me if I abbreviate mimeme to meme"

From *The Selfish Gene*, by Richard Dawkins (1976)

CASE STUDY: PEPE THE FROG

Pepe the Frog was created by American artist Matt Furie for his comic in 2005

Pepe was used in blog posts on Myspace and became an in-joke on Gaia Online

In 2008, the page containing Pepe and the catchphrase was scanned and uploaded to 4chan

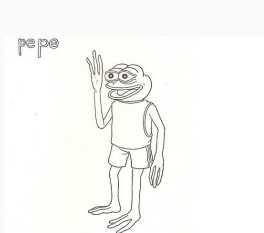


Figure: Original Pepe



Figure: Smug Pepe



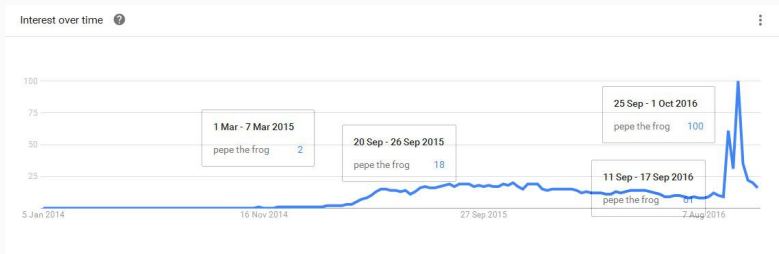
Figure: Angry Pepe

THREE IMPORTANT EVENTS

Around 2015, as Pepe's usage was increasing, a phenomenon began on **4chan** where users would declare certain variants as rare, known as a "rare Pepe"

In May 2016, a famous reporter wrote a couple of controversial article that raised hype among the people

In September 2016, an article published on Hillary Clinton's campaign website described Pepe as "a symbol associated with white supremacy" and denounced Donald Trump's campaign for its supposed promotion of the meme



CASE STUDY: PEPE THE FROG



Figure: Donald Trump's tweet

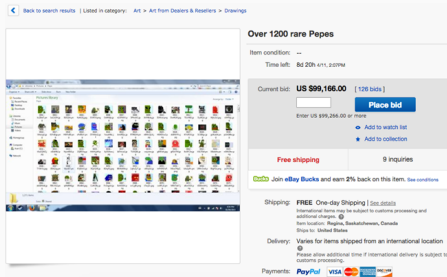


Figure: An interesting e-Bay post

In the initial phase of the spread of a particularly popular meme, a spike is seen in the number of people *infected* by it, but the number dies down once the meme is past its peak, the rate of decay depending on how fast people lose interest in the particular meme.

Another aspect of this model is that once someone loses interest in a meme, they might re-gain interest in it if some particularly interesting event leads to new implementations of the meme, which is similar to the concept of re-infection

The similarity hints towards a possibility of memetic modeling using the SIR model for pathogens.

MOTIVATION

The motivation behind the study of memetic modelling is derived from the fact that the internet today greatly reflects the human society

A study of the dynamics of memes can, thus, lead to a predictive model for estimating what the public might find interesting in future

Internet memes, as a subset of all memes, have a very special property - that of going "viral" at an extremely rapid rate, in a manner very similar to infectious pathogens



STANDARD SIR MODEL

The SIR model of viral propagation for a single pathogen is described in terms of three kinds of individuals in the population: the susceptible (S), the infected (I), and the recovered (R).

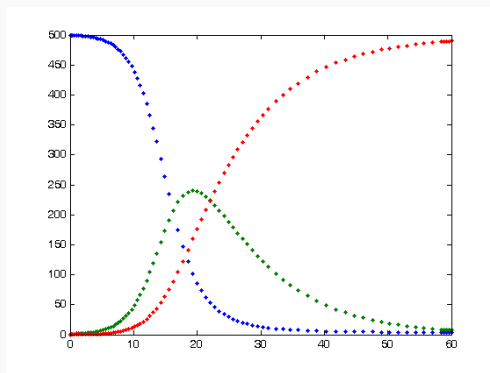


Figure: Blue=Susceptible, Green=Infected, and Red=Recovered

A basic model considers the population to be more or less constant at the relatively faster time-scale of viral propagation.

In our case, we re-define "susceptible" as all people connected to the internet who have not yet been exposed to the meme in consideration; "infected" as the people who have been exposed to the meme and still find the meme (or, its various implementations) interesting and, "recovered" as those people who have got bored of the meme, and are no longer interested in it.

The SIR model is described by the following differential equations:

$$\dot{S} = -\alpha SI$$

$$\dot{I} = \alpha SI + \beta IR - \gamma I$$

$$\dot{R} = \gamma I - \beta IR$$

Here α stands for the transmission rate proportionality constant and γ stands for the recovery rate proportionality constant. β denotes how much re-infection is prevalent

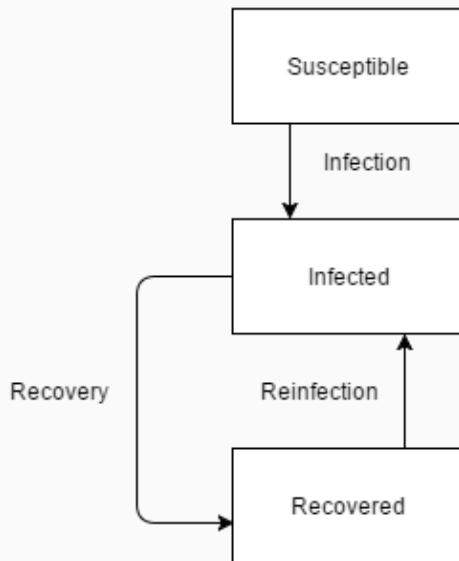
We have included β as the rate of re-infection of the *recovered* population. Here, $\beta < \alpha$, as argued earlier. There is however an implicit assumption in the model, as stated before, that the population $S + I + R = N$ remains constant.

$$\dot{S} = -\alpha SI$$

$$\dot{I} = \alpha SI + \beta IR - \gamma I$$

$$\dot{R} = \gamma I - \beta IR$$

$$N = S + I + R$$



STANDARD SIR MODEL

These are plots of numbers of people infected against the number who are susceptible (I vs S) (generated using *Mathematica*)

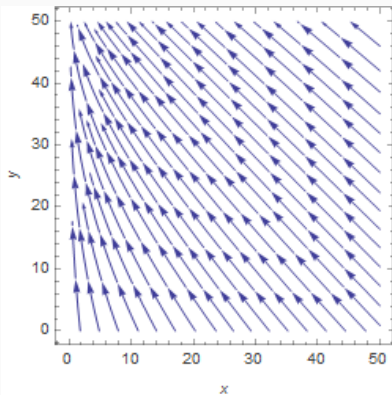


Figure: Phase diagram for the case where the recovery rate is small and the whole population ends up getting infected

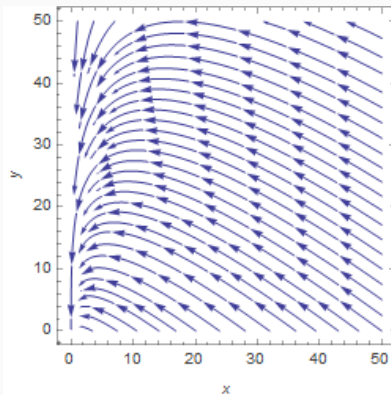


Figure: Phase diagram for the case where the recovery rate is sufficient to eventually result in all the people getting cured

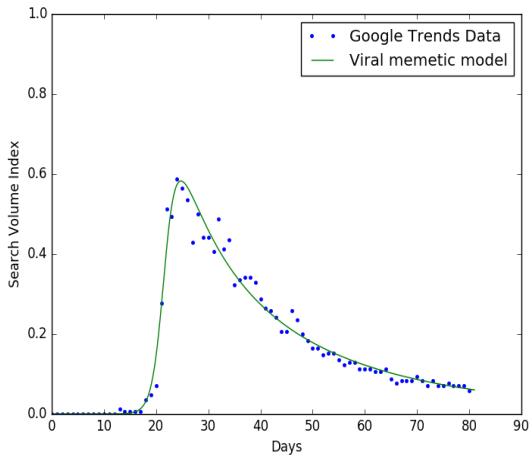
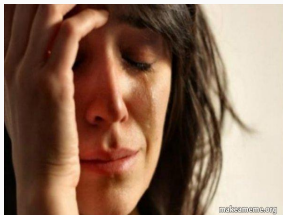
DEMONSTRATION OF DATA

All of the data is taken using Google Trends, a tool that gives the popularity of a given search term over time and geographical area

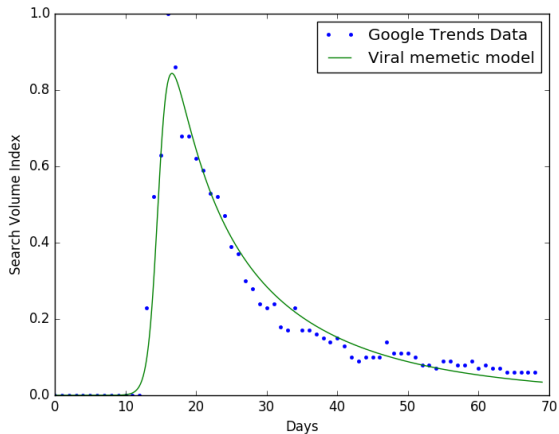


The curve fitting is done using SciPy

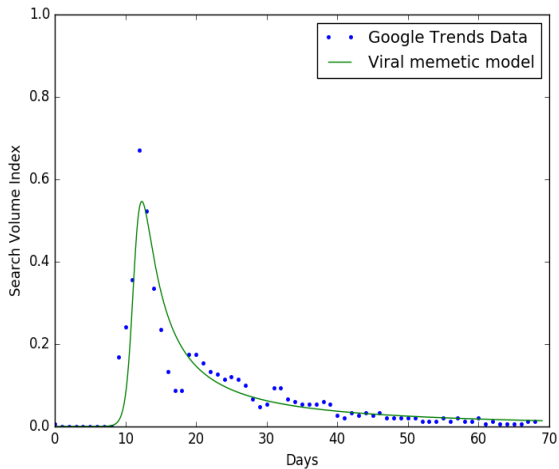
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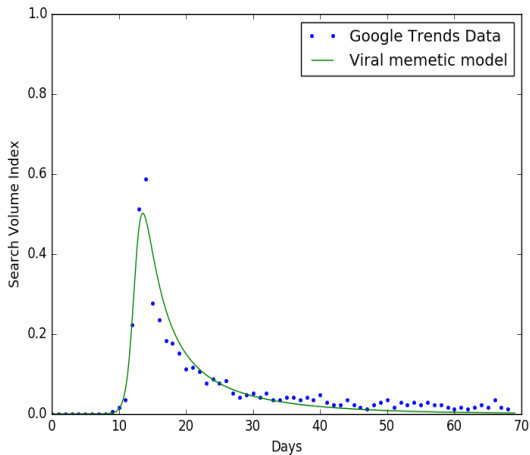
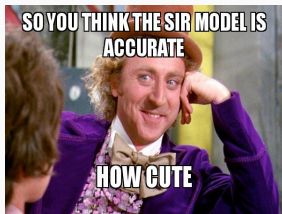
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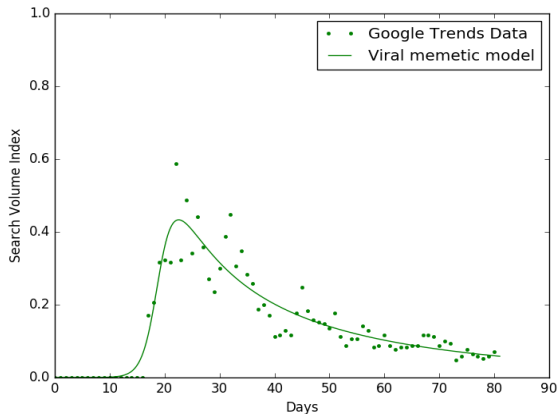
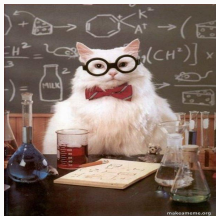
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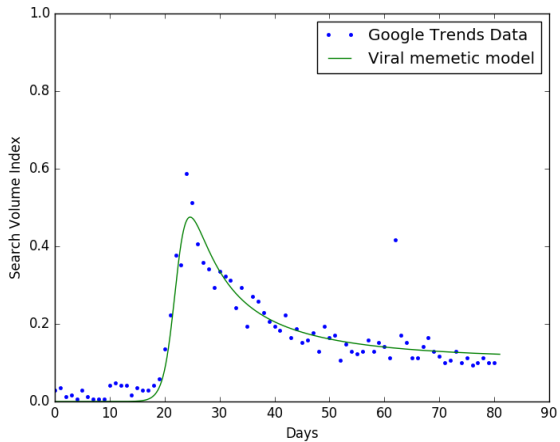
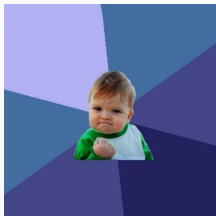
DEMONSTRATION OF DATA



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DEMONSTRATION OF DATA



EXTENSION TO MULTIPLE PATHOGENS

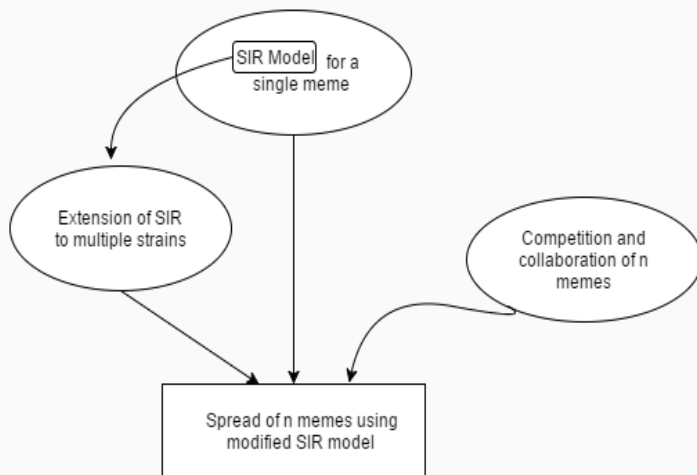


Figure: TODO: extension from one to many

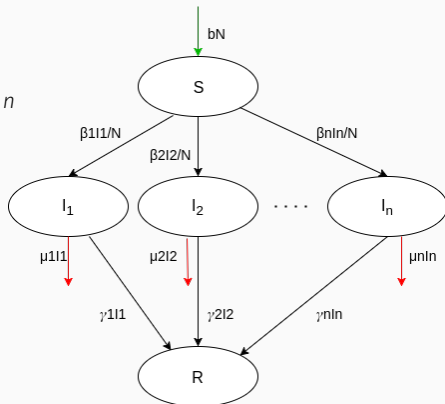
SIR MODEL FOR MULTIPLE PATHOGENS

$$\dot{S} = S \left(f(N) - \sum_{k=1}^n \beta_k \frac{I_k}{N} \right) + \sum_{k=1}^n b I_k + b R$$

$$\dot{I}_j = I_j \left(f(N) - b + \beta_j \frac{S}{N} - \gamma_j - \mu_j \right), \forall j \in 1, 2, \dots, n$$

$$\dot{R} = R(f(N) - b) + \sum_{k=1}^n \gamma_k I_k$$

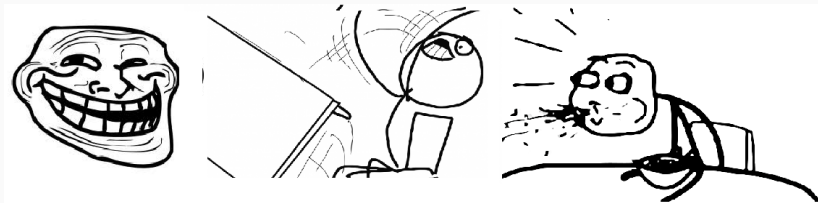
$$N = S + R + \sum_{k=1}^n I_k$$



Along with these, an overall per-capita death rate of $-d(N) = f(N) - b$

MULTIPLE SIR MODEL FOR MEMES

Almost all the internet memes are made for spread on social media and communication of an idea to people. Hence, since an internet meme actively influences the state of other memes as well, it is insufficient to study a single meme in isolation, without considering the effect of other memes.

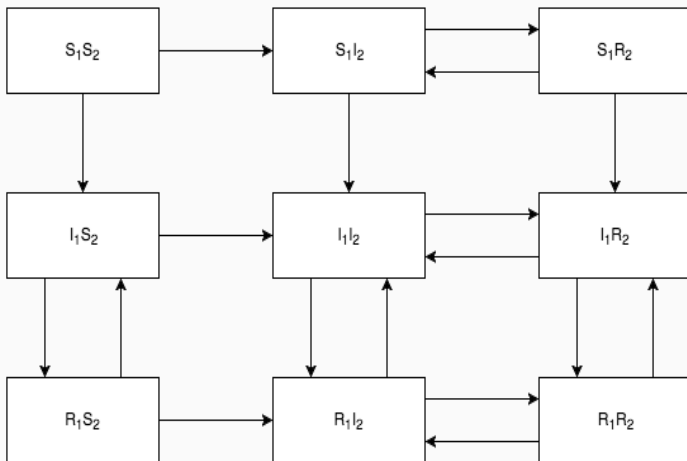


Here, we attempt to make a model for the spread of multiple memes on internet.

One important distinction that our model for memes has, as compared with the standard SIR model for pathogens, is that infection with one meme, does not rule out infection, with the other memes.

MULTIPLE SIR MODEL FOR MEMES

A person in the state S_1S_2 , in the next small time interval dt , can either remain in S_1S_2 , or move to either S_1I_2 or I_1S_2 . Similarly, from any category, a person can move to one of exactly 2 states, according to the diagram:



Writing down the 1st order differential equation, for change in any state, say S_1S_2 , will contain terms with change in state due to meme 1 and that due to meme 2.

Similarly, for N memes, the change in any state, can be broken down into changes in state due to each component meme. This fact, is the key step in being able to relate our model for multiple memes, to the single SIR model.

$$\frac{d}{dt}(S_1S_2) = \frac{\partial}{\partial t}(S_1S_2)|_{S_1S_2 \rightarrow I_1S_2} + \frac{\partial}{\partial t}(S_1S_2)|_{S_1S_2 \rightarrow S_1I_2}$$

Due to this property of each differential equation, we can see that we could actually work with memes 1 and 2 independently, and the solution could be written as the product of the two

So this model for two memes can be extended quite easily to N memes, with only the parameters of each meme dependent on other.

For j^{th} meme eliminating R_j by $R_j = 1 - S_j - I_j$

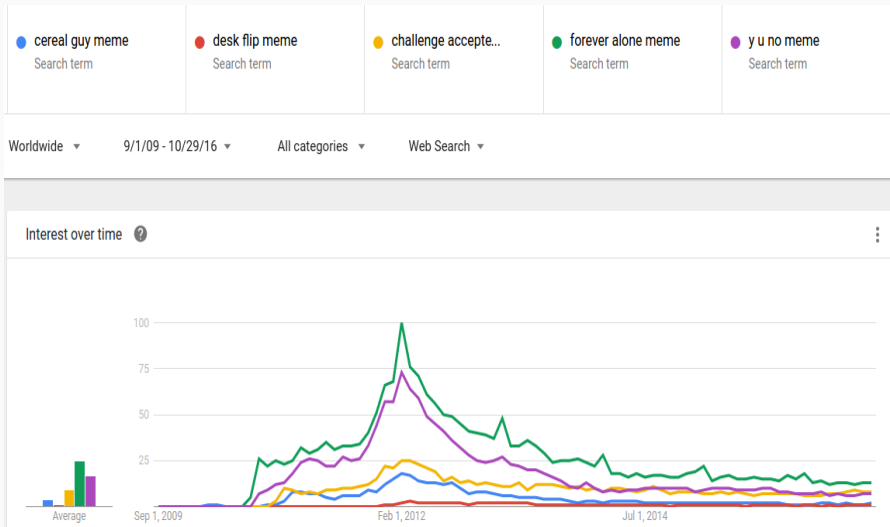
$$\frac{d}{dt}S_j = -\alpha_j S_j I_j$$

$$\frac{d}{dt}I_j = \alpha_j S_j I_j + \beta_j(1 - S_j - I_j)$$

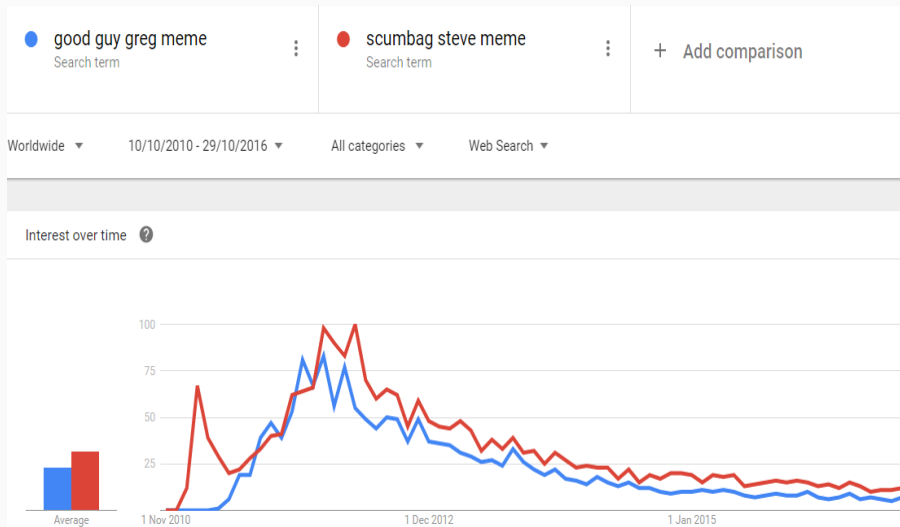
Eliminating t :

$$\frac{dI_j}{dS_j} - \frac{\beta_j}{\alpha_j} \frac{I_j}{S_j} = -\frac{(\alpha_j - \beta_j)}{\alpha_j} + \frac{(\gamma_j - \beta_j)}{\alpha_j S_j}$$

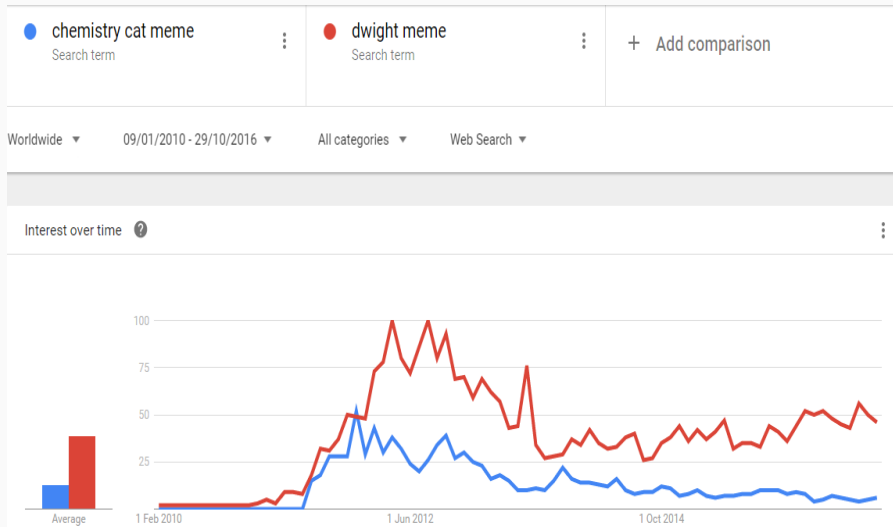
DEMONSTRATION OF COMPETITION AND COLLABORATION



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FAILURE OF SIR MODEL

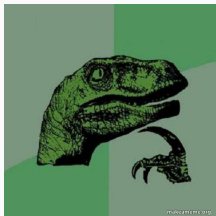
This simplified SIR model cannot handle multiple peaks in popularity of a meme, which is quite common in the real world, when we see new events leading to more, or equally successful implementations of the same general meme concept, as in the example of *Pepe the frog*.



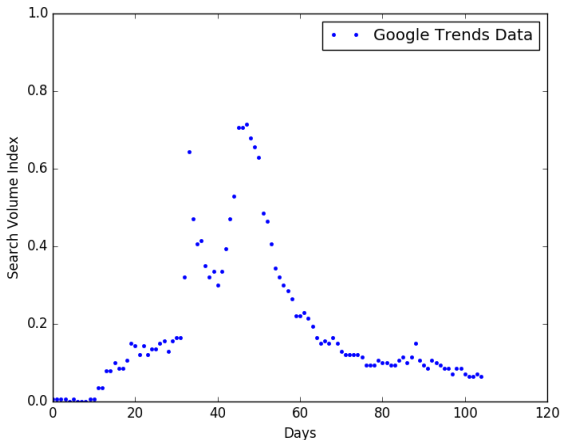
The second issue is with the fact that Google normalizes the Trends data in such a way that the highest index is assigned to the point which is the maximum in the concerned time span, thus removing any information about absolute search volume.

The fit function as a result is not accurate, and only shows the general trend of the meme, instead of actually making precise quantitative measurements.

FAILURE OF SIR MODEL



Here we have a quantitative example of the *Philoosoraptor* meme, where we couldn't fit the Google Trends data to our model.



We might have to consider different data mining tools for tackling problems like geographical spread of memes. Such a study would enable us to predict what kind of memes would be able to sustain in the public's mind, depending on the particular society they belong to.

The model can be made better by including additional degrees of freedom in order to allow multiple peaks in popularity.

- 1 Wang, Lin, and Brendan C. Wood. "An Epidemiological Approach to Model the Viral Propagation of Memes." *Applied Mathematical Modelling* 35.11 (2011): 5442-447
- 2 Ackleh, Azmy S., and Linda JS Allen. "Competitive exclusion in SIS and SIR epidemic models with total cross immunity and density-dependent host mortality." *Discrete Contin. Dyn. Syst. Ser. B* 5 (2005): 175-188
- 3 Coscia, Michele. "Competition and success in the meme pool: a case study on quickmeme. com." arXiv preprint arXiv:1304.1712